Quantum Mechanics I — Exercise Sheet 8

A particle on a ring of N sites

Variant: Questions only

Conventions, as in Lecture 8: sites e_x, $x \in \mathbb{Z}_N$, $N \geq 3$; the shift $Te_x = e_{x+1}$ and the mirror $Me_x = e_{-x}$; $H = c\mathbb{1} - \kappa(T + T^\dagger)$ with $\kappa > 0$; $\zeta = e^{2\pi i/N}$, $f_k = N^{-1/2} \sum_x \zeta^{kx} e_x$, $\varepsilon_k = c - 2\kappa \cos(2\pi k/N)$; $u = \kappa t/\hbar$. The particle starts on the site 0, and $a_x(t) = e^{ict/\hbar} \langle e_x, U_t e_0 \rangle$ is its amplitude on the site x without the common phase, so that $|a_x|^2$ is the probability of the site x.

Exercise 1 (three, four and five sites).

1. For $N = 4$, show that $a_0 = \cos^2 u$ and $a_2 = -\sin^2 u$. Where is the particle at $u = \pi/2$, and when is it first back on the site 0?
2. For $N = 3$, show that $a_0 = \frac{1}{3}(e^{2iu} + 2e^{-iu})$, and find the first $u > 0$ at which the particle is back on the site 0 with certainty.
3. For $N = 5$, find the two gaps $\varepsilon_k - \varepsilon_0$, and show that their ratio is irrational.

Exercise 2 (the average return).

1. Let $H = \sum_k \varepsilon_k E_k$ have distinct levels, and ψ be a unit vector with $w_k = \langle \psi, E_k \psi \rangle$. Show that the average over long times of $|\langle \psi, \psi_t \rangle|^2$ is $\sum_k w_k^2$.
2. Find this average for the particle started on the site 0 of the ring of six sites, and of four sites.

Exercise 3 (the flux at its special values). Let $H_\theta = c\mathbb{1} - \kappa(e^{i\theta}T + e^{-i\theta}T^\dagger)$ as in § 8.6, and for an integer r let W_r be the unitary with $W_r e_x = \zeta^{-rx} e_{-x}$.

1. Show that at $\theta = \pi r/N$, W_r commutes with H_θ, and $W_r f_k = f_{r-k}$.
2. Conclude that the levels of k and $r - k$ coincide there, and check it with the formula for $\varepsilon_k(\theta)$.

Exercise 4 (two sublattices). On a ring of $2N$ sites, $N \geq 2$, with $\zeta = e^{2\pi i/2N}$ and the cells of § 8.7, let the on-site energies alternate: $H = \Delta\Lambda - \kappa(T + T^\dagger)$, where $\Lambda e_y = (-1)^y e_y$ and $\Delta, \kappa > 0$.

1. Show that T is no longer a symmetry, but T^2 and M are. When do they commute?
2. Use Theorem 8.3 to find $h(k)$ and the levels of H.
3. Find the gap between the two bands, for even and for odd N, and its limit as $\Delta \to 0$.
4. For $N = 2$ (four sites), list the levels. Is any of them degenerate?

Exercise 5 (⋆ which rings return). Show that the particle started on the site 0 comes back to it exactly, at some $u > 0$, only for $N = 3$, 4 and 6.